

# Yichao Jin, Ph.D.

School of Economic, Political and Policy Sciences  
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*Specialization: Behavioral AI / Health Economics / Behavioral Economics / Discrete Choice Experiments*

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## Education

- University of Texas at Dallas** May 2026  
**Ph.D. in Public Policy and Political Economy**  
- Successfully Defended: April 2026  
- Dissertation: “Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan”  
- Committee: [Dohyeong Kim](#) (Chair), [Richard Scotch](#), [Soyoung Kwon](#), [Khai Chiong](#).  
- Distinction: Dean of Graduate Education Dissertation Research Award (2025).
- University of Texas at Dallas** May 2026  
**M.S. in Social Data Analytics and Research**
- University of Queensland** 2019  
**Master of Development Economics**
- University of California, Riverside** 2017  
**B.A. in Economics**

## Research Fields

**Primary:** Health Economics, Applied Econometrics, Behavioral Economics, Behavioral AI  
**Secondary:** Quantitative Methods, Policy Evaluation, Technology Governance (AI in Health)

## Working Papers

1. “Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust”  
*Job Market Paper* — [Link to Full Draft](#)
  - Developed a structural Discrete Choice Experiment (DCE) framework to analyze the interaction between temporal barriers and institutional trust in vaccination behavior.
  - Identified nonlinear heterogeneity in waiting time sensitivity, providing a behavioral explanation for vaccine hesitancy.
2. “Empirical-Frontier Regularization for LLM Synthetic Agents: A Behavioral Digital Twin Framework”  
*Working Paper* — [SSRN Link](#). Submitted to AI and Economics (AIE) Summer Conference 2026. Developed a 3-stage framework integrating DCE data and Structural Causal Modeling (SCM) to align generative agents with empirical choice frontiers, achieving a 34% reduction in frontier-deviation MSE.

3. **“Ideological Contamination in Policy Simulation: Separating Political Bias from Model Capabilities in Predicting Administrative Burden Effects”**  
*Working Paper* — Investigates the disentanglement of inherent LLM ideological priors from structural predictive capabilities. Proposes a bias-correction methodology within the BDT framework to enhance the fidelity of administrative burden simulations.
4. **“The Ghost in the Machine: Algorithmic Rationality vs. Social Legitimacy in AI-Driven Public Health Governance”**  
*Working Paper* — Analyzes the theoretical tension between automated efficiency and institutional trust using the AGIL framework and Rational Choice models.
5. **“The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount”**  
*Manuscript in Preparation (Expected submission: August 2026)* — Targeting *Social Science & Medicine*.

## Research Experience & Affiliations

### Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.

### Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

## Teaching Experience

### Teaching Assistant, School of Economic, Political and Policy Sciences, UT Dallas

- **Quantitative Methods for Policy Analysis (Graduate)**: Led Econometrics labs; specialized in R programming, regression analysis, and causal inference.
- **Health Economics & Public Policy (Undergraduate)**: Developed case studies on health market failures and administrative challenges.
- **Public Policy Analysis (Undergraduate)**: Mentored students in drafting evidence-based policy memos.

## Conference Presentations

- **AI and Economics (AIE) Summer Conference 2026**: Empirical-Frontier Regularization for LLM Synthetic Agents.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **UT Dallas Graduate Research Symposium 2025**: Modeling Vaccination Decisions with Hyperbolic Discounting.

## Honors & Awards

- **2025 Dean of Graduate Education Dissertation Research Award**, UT Dallas

- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

## Technical Skills & Certifications

**Econometrics:** Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.

**Software:** Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.

**Certifications:** Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

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## REFERENCES

### Dohyeong Kim (Chair)

University of Texas at Dallas

Senior Associate Dean of Graduate Education

Robert E. Holmes Jr. Professor of Public Policy,

GIS & Social Data Analytics

Team Lead, SHARP

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### Soyoung Kwon

Associate Professor of Sociology and Public Health

Editorial Board, *Society and Mental Health*

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### Richard Scotch

Professor Emeritus of Sociology,

Public Health, and Public Policy

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